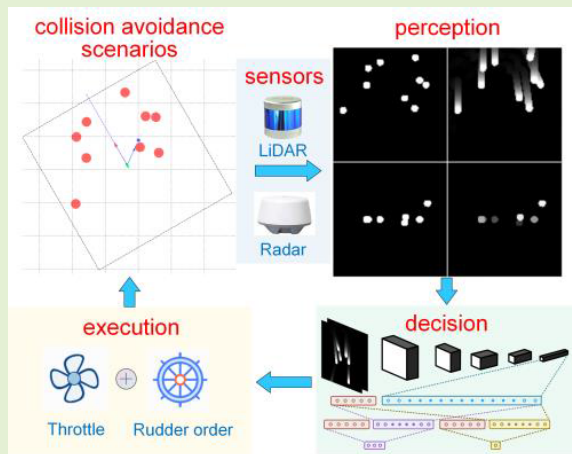
Research on Collision Avoidance Algorithm of Unmanned Surface Vehicle Based on Deep Reinforcement Learning

Jiawei Xia, Xufang Zhu, Zhikun Liu, Yasong Luo, Zhaodong Wu, and Qiuhan Wu

***Abstract*—To solve the problem of active collision avoidance for unmanned surface vehicles (USVs) in a complex maritime environment, a method of deep reinforcement learning (DRL) based on proximal policy optimization (PPO) is proposed. To master the collision avoidance policy by self-learning, the mathematical model of USV, dynamic obstacle generation model, and reward mechanism were established. Using the relative position between the obstacle and USV and the information of the closest point of approach (CPA), high-dimensional state features including the collision track layer and collision threat layer were constructed. On this basis, combined with low-dimensional states such as navigation state and path error, a deep convolutional neural network (CNN) structure fused in multifeature and multiscale was designed. The proposed DRL network was trained through repeated collision avoidance simulations in random environments. Simulation results reveal that the proposed algorithm can output effective decisions in complex scenarios and avoid dynamic obstacles quickly and safely.**



***Index Terms*—Collision avoidance, convolutional neural network (CNN), deep reinforcement learning (DRL), proximal policy optimization (PPO), unmanned surface vehicle (USV).**

I. Introduction

AS A small platform for performing surface tasks, unmanned surface vehicles (USVs) have the advantages of being high speed, intelligent, flexible, concealing, low cost, and causing no casualties, which has great value both in the military and civil use [1]. With the development of wireless communication and navigation control technology, USVs have made important breakthroughs such as path tracking and formation coordination control [2], [3], [4], [5]. The USV's autonomous collision avoidance capability is key to ensuring its safe navigation and successful mission execution in complex maritime environments.

According to the task hierarchy, collision avoidance problems can be divided into global path planning collision avoidance and local collision avoidance. Global path planning methods generate continuous paths to avoid collisions according to task requirements, which are based on known obstacles map information in the sea area. Local path planning methods usually generate avoidance paths temporarily based on the real-time sensor information when the collision threat is detected and restore the original navigation state after the threat disappears. This article focuses on local collision avoidance.

Conventional local collision avoidance algorithms can be divided into two categories according to the technical route. One category is according to geometric rules, which simplifies the obstacles and ships to a convex body while considering kinematic constraints and then calculates the upper and lower boundaries of a feasible path. The typical algorithms include the velocity obstacle method [6], [7], [8], artificial potential

Manuscript received 26 August 2022; revised 12 October 2022; accepted 10 November 2022. Date of publication 28 November 2022; date of current version 31 May 2023. This work was supported in part by the Natural Science Foundation of Hubei Province of China under Grant 2018CFC865 and in part by the China Postdoctoral Science Foundation Funded Project under Grant 2016T45686. The associate editor coordinating the review of this article and approving it for publication was Dr. Pu (Perry) Wang. *(Corresponding author: Xufang Zhu.)*

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Digital Object Identifier 10.1109/JSEN.2022.3222575

field (APF) method [5], [9], [10], [11], and dynamic virtual ship method [12], [13]. However, the methods above may trap into local optimum or no solution when confronted with a complex environment. The other category is according to path search, which searches out a reasonable path through optimizing, such as the dynamic window method [14], [15], A* algorithm, rapid expanding random tree, and other graph search algorithms [16], [17], [18], [19]. The path searching method is only applicable to avoid static obstacles or obstacles of low relative speed and has higher computational complexity, which is difficult for high-speed USVs to realize real-time collision avoidance.

With the progress of deep reinforcement learning (DRL) technology, a large number of research achievements have emerged in the field of AI gaming, automatic driving, robot control, and so on [20], [21], [22], [23], [24], [25], [26]. This kind of algorithm interacts with the environment extensively through the agent and learns and optimizes policy from trial and error. It has a strong adaptive ability in the complex environment, which provides a new solution to the collision avoidance problem. For local collision avoidance of USVs, a deep Q network (DQN) is used to achieve USV collision avoidance in fixed scenarios [27]. Ma et al. [28] used the DDPG algorithm to deal with the collision avoidance problem between USV formations. However, the state input dimension of the deep network is related to the number of USV formations, which does not involve policy for dealing with noncooperative targets. Wu et al. [29] studied the dynamic collision situation and used a grid map image as state input of the deep network to make USV successfully cross the collision area, but the navigation path constraint was not considered. Woo and Kim [30] introduced the semi-Markov decision model and designed a navigation policy switcher based on DQN. It can switch between three navigation modes according to the ambient state of the USV, involving port-side collision avoidance, starboard-side collision avoidance, and keeping a straight line. The effectiveness of collision avoidance has been proven by maritime trials, but the control process and implementation mechanism of this method are relatively complicated.

To solve the collision avoidance problem of USVs in the complex navigation environment and make full use of the representation ability of the DRL deep network, an algorithm for collision avoidance of USVs based on proximal policy optimization (PPO) is proposed in this article, which is a model-free, end-to-end path planning algorithm. It senses high-dimensional obstacle information and low-dimensional state information of USVs in real-time through a deep convolutional neural network (CNN) fused with multifeature and multiscale outputs policy and guides USVs to reach the destination along the established route while meeting collision avoidance requirements. Compared with the traditional collision avoidance algorithm, our method has the characteristics of adaptability and versatility.

Salient contributions in this article can be summarized in the following.

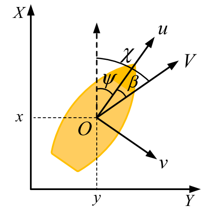

Fig. 1. Coordinate system of a USV.

1) Combined with DRL, USVs learn to avoid obstacles autonomously in an end-to-end manner, making decisions based on sensor input directly.
2) The situation of collision avoidance is represented in the form of multichannel pictures, and the environment features are extracted and encoded by CNNs.
3) Two novel layer features were proposed, including an obstacle track layer representing the historical track and a collision threat layer representing future prediction based on the distance to the closest point of approach (DCPA) and time to the closest point of approach (TCPA).

The remainder of this article is organized as follows. In Section II, the USV dynamic model, the background and the process of collision avoidance are presented. In Section III, each component of the proposed DRL collision avoidance algorithm is given in detail, including the reinforcement learning (RL) algorithm, environment design, reward function design, state space design, and deep network design. In Section IV, the DRL-based collision avoidance algorithm is trained and compared. Finally, the conclusions of the research and further discussion are reviewed in Section V.

II. USV Modeling and Collision Avoidance Process

A. Underactuated USV Modeling

Generally, a three-degree-of-freedom (3-DoF) mathematical model is established to study the underactuated USV [31]. The coordinate system used in this work is shown in Fig. 1. Define $\boldsymbol{\eta} = [x, y, \psi]^{\mathrm{T}}$ as the horizontal position and heading angle of USVs in the earth-fixed inertial frame. Define $\boldsymbol{\upsilon} = [u, \upsilon, r]^{\mathrm{T}}$ as the velocity and angular velocity in the body-fixed reference frame.

The equation of USV's 3-DoF kinematic model is

$$\dot{\boldsymbol{\eta}} = \boldsymbol{R}(\boldsymbol{\eta})\,\boldsymbol{\upsilon} \tag{1}$$

where $\boldsymbol{R}(\boldsymbol{\eta})$ represents the transformation matrix from the body-fixed frame into the earth-fixed inertial frame

$$\boldsymbol{R}(\boldsymbol{\eta}) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The equation of USV's 3-DoF dynamic kinematic model is

$$\boldsymbol{M}\dot{\boldsymbol{\upsilon}} + \boldsymbol{C}(\boldsymbol{\upsilon})\,\boldsymbol{\upsilon} + \boldsymbol{D}(\boldsymbol{\upsilon})\,\boldsymbol{\upsilon} = \boldsymbol{\tau}_c + \boldsymbol{\tau}_e \tag{2}$$

where $\boldsymbol{M}$ represents the inertial matrix, $\boldsymbol{C}(\boldsymbol{\upsilon})$ denotes the Coriolis and centripetal terms, $\boldsymbol{D}(\boldsymbol{\upsilon})$ is the nonlinear damping

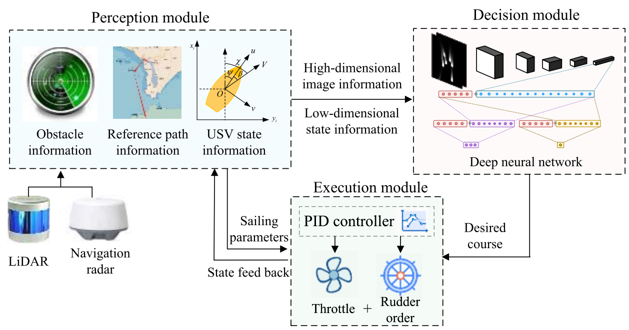

Fig. 2. Schematic of the collision avoidance system based on DRL.

matrix; $\boldsymbol{\tau}_c = [\tau_u, 0, \tau_t]^{\mathrm{T}}$, τ_u is the surge force, τ_t is the yaw moment, $\boldsymbol{\tau}_e = [\tau_{eu}, \tau_{ev}, \tau_{er}]^{\mathrm{T}}$, and τ_{eu}, τ_{ev}, and τ_{er} are the environmental interference of the USV in the direction of u, v, and r, respectively,

$$\boldsymbol{M} = \begin{bmatrix} m_{11} & 0 & 0 \\ 0 & m_{22} & m_{23} \\ 0 & m_{32} & m_{33} \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 0 & 0 & c_{13} \\ 0 & 0 & c_{23} \\ -c_{13} & -c_{13} & 0 \end{bmatrix},$$

$$\boldsymbol{D} = \begin{bmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & d_{23} \\ 0 & d_{32} & d_{33} \end{bmatrix}, \quad c_{13} = -m_{22}v - \frac{1}{2}(m_{23} + m_{32})\, r,$$

$c_{23} = m_{11}u$.

Equation (2) can be simplified as

$$\begin{cases} \dot{u} = \dfrac{1}{m_{11}}\left(m_{22}ur + \dfrac{1}{2}(m_{23} + m_{32})\, r^2 - d_{11}u + \tau_u + \tau_{eu}\right) \\ \dot{v} = \dfrac{1}{m_{22}}\left(-m_{23}\dot{r} - m_{11}ur - d_{22}v - d_{23}r + \tau_v + \tau_{ev}\right) \\ \dot{r} = \dfrac{1}{m_{33}}\Big(-m_{32}\dot{v} - m_{22}vu - \dfrac{1}{2}(m_{23} + m_{32})\, ru + m_{11}uv \\ \qquad\qquad - d_{32}v - d_{33}v - d_{33}r + \tau_r + \tau_{er}\Big). \end{cases}$$

B. Process of Collision Avoidance

In view of real-time collision avoidance requirements in complex waterways or areas with intensive fishing boats, the collision avoidance process can be described as follows: USVs start from a given starting point and navigate along the reference path. They use perception sensors such as radar and LiDAR to perform obstacle detection. When they obtain information about obstacles and perceive the threat of collision, they make decisions and actively change course to perform collision avoidance actions. After the threat is eliminated, they resume their original course.

The schematic of the collision avoidance system based on DRL proposed in this article is shown in Fig. 2.

The collision avoidance system consists of a perception module, a decision module, and an execution module. The perception module is used to summarize and process the information on the obstacles, reference path, and USV state, and then convert them into high-dimensional image information and low-dimensional state information. The decision module is composed of the deep neural network, which outputs the desired course according to the state information. The execution module is used to output the control quantity of throttle and rudder order according to the current sailing parameters and desired course and then update its status information according to the USV dynamic model and kinematic model. The three modules above are executed recurrently to realize USV dynamic real-time collision avoidance.

III. Collision Avoidance Algorithm Based on DRL

This section describes the technical details involved in the implementation of the proposed DRL collision avoidance algorithm. First, it introduces the learning algorithm required for collision avoidance, then designs the rules of the DRL interactive environment and collision avoidance problem, and finally defines the reward function, state space, and structure of the deep neural network.

A. DRL Algorithm

RL belongs to the machine-learning domain, which is an important solution to the Markov decision process (MDP) problem. DRL is built based on RL and has the ability to solve complex decision problems by combining it with the deep neural network. In recent years, multiple efficient DRL algorithms have been proposed with the intensive study on DRL, such as Actor-Critic [32], A3C [33], DDPG [34], PPO [35], and so on. PPO is a stochastic policy gradient algorithm based on the Actor-Critic framework, which originates from the trust region policy optimization algorithm [36]. On this basis, it improves the learning performance and simplifies the computation by improving the objective function. In this article, the discrete action space version of PPO is used as the learning algorithm of DRL.

Define the action chosen by USVs in the state s as a and the policy π is modeled as $\pi(a|s)$, which represents the probability of action a to be chosen by the policy π in the state s.

The policy π is a deep policy network in DRL, and the parameters of the network are denoted as θ. The state of the system at the moment t is denoted as s_t, the deep policy network π_θ inputs s_t and outputs the probability of each possible action, then the action of USVs a_t at the moment t is obtained through probability sampling. At the same time, the environment will give rewards $r_{t+1}(s_t, a_t)$ based on s_t and a_t. The objective of RL is to produce the optimal policy π^* that maximizes the cumulative discount reward R_t

$$R_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \gamma^3 r_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k r_{k+1}. \tag{3}$$

In (3), γ is the discount factor. The smaller the value of γ is, the more attention will be paid to maximum short-term rewards, and conversely the long-term rewards.

The policy π can be evaluated by the state value function $V^\pi(s)$ and the action value function $Q^\pi(s, a)$. $V^\pi(s)$ represents the desired reward of the intelligent agent making decisions according to policy π from the state s. $Q^\pi(s, a)$ represents the desired reward of all possible decision sequences after action a is executed according to the policy π from the state s

$$V^{\pi}(s_t) = \mathbb{E}_{\pi}[R_t|s_t] = \mathbb{E}_{\pi}\left[\sum_{k=0}^{\infty}\gamma^k r_{k+1}|s_t\right] \quad (4)$$

$$Q^{\pi}(s_t, a_t) = \mathbb{E}_{\pi}[R_t|s_t, a_t] = \mathbb{E}_{\pi}\left[\sum_{k=0}^{\infty}\gamma^k r_{k+1}|s_t, a_t\right]. \quad (5)$$

Define $A^{\pi}(s_t, a_t)$ as the advantages achieved by the USV when performing an action a in the state s compared to other actions

$$A^{\pi}(s_t, a_t) = Q^{\pi}(s_t, a_t) - V^{\pi}(s_t). \quad (6)$$

Define $\hat{A}_t$ to represent the estimation of the advantage function $A^{\pi}(s_t, a_t)$ at the time t. T represents the number of time steps for the continuous implementation of the policy π in an episode, and $\hat{A}_t$ is

$$\hat{A}_t = -V^{\pi}(s_t) + r_t + \gamma r_{t+1} + \cdots + \gamma^{T-t+1} r_{T-1} + \gamma^{T-t} V^{\pi}(s_T). \quad (7)$$

The policy gradient method is an RL algorithm based on policy search. This method calculates the estimated value of the policy gradient and applies it to the stochastic gradient ascent algorithm. The policy gradient loss function $L^{\mathrm{CLIP}}(\theta)$ is designed as

$$r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \quad (8)$$

$$L^{\mathrm{CLIP}}(\theta) = \hat{\mathbb{E}}_t\left[\min\left(r_t(\theta)\hat{A}_t, \mathrm{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t\right)\right]. \quad (9)$$

In (8), $r_t(\theta)$ is the probability ratio. In (9), ε is the hyperparameter, and the clip function is utilized to limit $r_t(\theta)$ within the interval $[1-\varepsilon, 1+\varepsilon]$.

The PPO network generally adopts the framework of Actor-Critic, including both the policy network π_{θ} and the value network V_{ϕ}. π_{θ} inputs state feature $\boldsymbol{S}$ and outputs probability distribution of action a. The input of V_{ϕ} is also $\boldsymbol{S}$, which outputs the value prediction $V_{\phi}(\boldsymbol{S})$ of the state $\boldsymbol{S}$. During training, the gradient ascent method is used to update the network parameters, so that the loss function can be minimized as much as possible, and the agent can gradually master the collision avoidance policy.

B. Design of Collision Avoidance Environment

A virtual collision avoidance environment is an important part of RL, which should simulate all probable complex maritime situations as much as possible and also reflect diversified task styles and sailing experiences. This section defines the task rules of the environment, the principle of random obstacle generation, and the guidance method of the reference path.

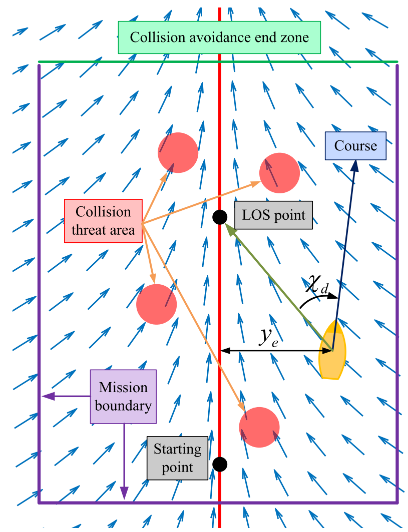

Fig. 3. Schematic of the collision avoidance scenario.

1) Design of Task and Action Space: The schematic of the collision avoidance scenario is shown in Fig. 3. The USV navigation area is defined as the rectangular sea area, with the red reference path as the center and extending to both sides as the mission area. The boundary of the mission area along the direction of the reference path is the collision avoidance end zone, and the red circle is the collision threat area. The USV starts from the starting point and drives along the red reference path until it reaches the collision avoidance end zone safely, which is regarded as a successful mission. During navigation, the USV must not enter the collision threat area or go beyond the mission boundary; otherwise, it will be regarded as collision avoidance failure. At the beginning of each time step, an action needs to be selected from the action set $\mathbb{A}$ to update the desired course ψ_d. The designed action set $\mathbb{A} = \{-\Delta\psi_d, 0, \Delta\psi_d\}$ contains three kinds of control instructions, which are to increase/decrease the desired course by a certain degree or to keep the current course unchanged. $\Delta\psi_d$ depends on the steering capability of the USV. The DRL policy network will generate the selection probabilities for the three actions described above. In the network training stage, an action is randomly selected from $\mathbb{A}$ according to the probability distribution to ensure that the action selection is random and exploratory.

2) Design of Obstacle: An indefinite number of obstacles with uniform linear movement are generated in the environment, which can be divided into high-threat obstacles and low-threat obstacles according to the threat level. The DCPA between obstacles and the USV on the desired course is used to determine the threat level. If the DCPA between an obstacle and the USV at the initial moment is less than the radius of the threat zone of the obstacle, the obstacle is a high-threat obstacle; otherwise, it is a low-threat obstacle. The purpose of collision threat level division is to generate targeted scenarios that may cause collisions so that the DRL network can experience and deal with collision threat-related decisions as much as possible, so as to accelerate network convergence. The threat level, position, and motion parameters of collisions are randomly generated after the environment reset to ensure the complexity of the collision avoidance environment.

3) Design of Guidance Policy: The line-of-sight (LOS) method is adopted to map the desired position (x_d, y_d) to the LOS angle ψ_{los}. LOS guidance policy is a classic and effective navigation algorithm [37]. The green arrow in Fig. 3 represents the direction calculated by LOS at the corresponding position. By introducing the guidance policy, the distance error y_e and course error $\chi_d = \psi_{\text{los}} - \psi$ are provided for the RL algorithm, thus providing input for eliminating the position deviation between the USV and the reference path.

C. Design of Reward Function

The article designed a suitable reward function to evaluate the current state of USVs in real-time to meet USV path tracking and real-time collision avoidance at once. The design of the reward function is expected to follow these rules: the closer to the reference path the USV is, the closer to the desired course the course is, and the more stable the course is, the higher the reward value will be. During the navigation of USVs, the designed reward function is composed of position error reward r_d, course error reward r_ψ, and course stability reward r_s

$$\begin{cases} r_d = 2e^{-k_1|y_e|} - 1 \\ r_\psi = 2e^{-k_2|\chi_d|} - 1 \\ r_s = 2e^{-k_3|\sigma|} - 1. \end{cases} \tag{10}$$

In the above equation, the negative exponential function is adopted to describe rewards, and the range of each subreward is limited to $(-1, 1]$. k_1, k_2, and k_3 are the adjustment coefficients of each subreward, and σ represents the standard deviation of USV heading in a period of time. By introducing course stability reward r_s, DRL can be inclined to form a continuous and stable course control policy during training, thus accelerating convergence. The reward function in the navigation process is defined as

$$r = w_d r_d + w_\psi r_\psi + w_s r_s. \tag{11}$$

In the above equation, w_d, w_ψ, and w_s are the weight coefficients.

When USVs go out of boundaries or a collision occurs, the task terminates and generates termination rewards. According to different situations, the rewards obtained are designed as follows: when USVs go out of boundaries, $r = r_o$, r_o represents the penalty reward of going out of boundaries, and the value is negative. When USVs collide with an obstacle, $r = r_c$. Similarly, r_c represents the penalty reward of collision, and the value is negative. When USVs reach the collision avoidance end zone, $r = r_e e^{-k_4|y_e|}$, where $k_4$is the adjustment coefficient, r_e is the completion reward, and the value is positive. The closer to the reference path is the USV when it reaches the end zone, the higher the termination reward.

D. Design of State Space

To design the state space of the USV collision avoidance problem, it is necessary to consider how to effectively extract the position and motion features of collisions as well as the state of the USV itself, so that the DRL network can obtain sufficient environmental information to make correct decisions. According to the state dimension, the state features of the DRL network input are divided into a high-dimensional state and a low-dimensional state.

1) Design of High-Dimensional State Feature: The high-dimensional state features mainly represent the situation information related to collisions through 2-D state data. In [14] and [15], the collision motion parameter list is used as the high-dimensional state features. However, this method is limited by a certain number of obstacles and requires USVs to have complete situation awareness ability, so it is difficult to be applied in the dynamic collision avoidance environment. Compared with the target parameter list, the grid map image information can contain more information [16], [17]. The USV usually uses sensors such as LiDAR or navigation radar to perceive close-range obstacle situations, its observation range of the environment is limited, and it cannot master the global obstacle information. To solve this problem, a rectangular situation sense area (936 × 936 m) centered on the USV and fixedly connected to the hull is constructed as the representation area of the high-dimensional state feature [30], assuming that the information of all obstacles in this area is known. In this article, a grid map image (117 × 117 pixels) is selected as the input form of the DRL high-dimensional state feature, the obstacle track layer is constructed to represent the track information of obstacles relative to the USV and the collision threat layer to represent the predicated position and threat level of collision.

a) Obstacle track layer: Denote the number of obstacles in the situation awareness area observed by the USV at time t as n. Define in the body-fixed reference frame $\{b\}$ the position of the ith obstacle is $\boldsymbol{\eta}_i^t = [x_i^t, y_i^t]^{\mathrm{T}}$ and threat radius of which is r_i, $i = 1, 2, \ldots, n$. Set up the layer coordinate frame $\{l\}$ with the pixel in the center of the layer as the origin point, the x-axis horizontally to the right, and the y-axis vertically upward.

Define the mapping matrix that transforms the body-fixed reference frame $\{b\}$ into layer coordinate frame $\{l\}$ as $\boldsymbol{R}(\kappa) = \text{diag}(\kappa, \kappa)$, where κ is the mapping coefficient, represents the number of pixels of layer coordinate frame $\{l\}$ in the unit distance of the body-fixed reference frame $\{b\}$, the value of κ in this article is 1/8 pixel/m for navigation radar.

Define the position layer $\boldsymbol{L}_p^t$ to represent the position of the obstacle in the layer coordinate frame $\{l\}$. According to the mapping relationship, the coordinates and radius of the location feature of the obstacle in $\{l\}$ at the time t are, respectively, $\boldsymbol{R}(\kappa)\boldsymbol{\eta}_i^t$ and κr_i, respectively, and set the intensity value of the area covered by all the obstacles in $\boldsymbol{L}_p^t$ is 255. Define the obstacle track layer as $\boldsymbol{L}_s^t$, and its recursive expression is

$$\begin{cases} \boldsymbol{L}_s^t = \text{clip}\left(\gamma_s \boldsymbol{L}_s^{t-1} + \boldsymbol{L}_p^t, 0, 255\right) (t > 1) \\ \boldsymbol{L}_s^t = \boldsymbol{L}_p^t \ (t = 1) \end{cases} \tag{12}$$

where γ_s is the decay coefficient of position. The function clip is used to limit the pixel intensity in the layer within the interval $[0, 255]$. The purpose of this equation is to generate

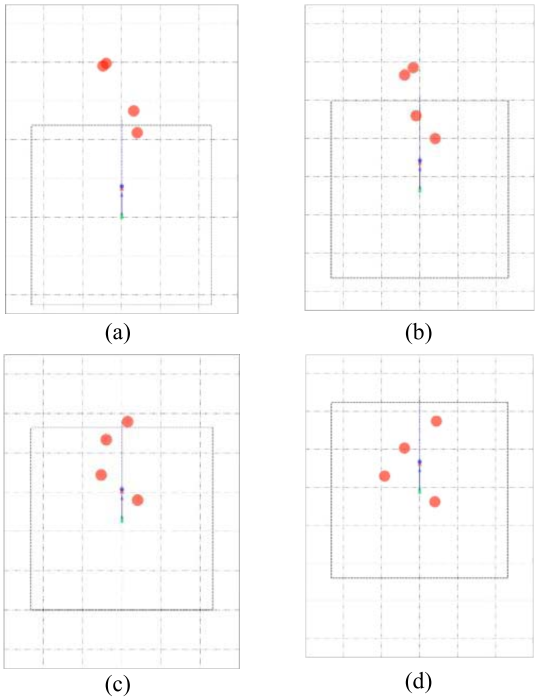

Fig. 4. Schematic of obstacle avoidance scenarios in different moments. (a) $t = 55$. (b) $t = 80$. (c) $t = 105$. (d) $t = 130$.

the track features of obstacles relative to the USV in the track layer.

Fig. 4 shows the collision avoidance scenario of the USV in different moments, which simulates USV driving along the reference path and crossing four dynamic obstacles. Each red circle represents a dynamic obstacle, and the dotted rectangle represents the situation awareness area of the USV. Fig. 5 shows the position layer image at the corresponding moment, which is consistent with the obstacle situation in the situation awareness area. Fig. 6 shows the layer image of the obstacle track. It can be seen that the intensity of the historical track of the obstacle gradually decreases with the accumulation of time, achieving a display effect similar to radar afterglow. Therefore, this layer can represent the location feature and historical track of the obstacle relative to the USV.

b) Collision threat layer: The obstacle track layer $\boldsymbol{L}_s$ provides past and present features of the obstacle, while the collision threat layer provides predictions of future collision threats. In this section, a collision threat layer construction method based on DCPA and TCPA is proposed by calculating the location relation and time margin between each obstacle and the USV in the situation awareness area.

In the body-fixed reference frame $\{b\}$, the DCPA of the ith obstacle relative to the USV is d_i, the TCPA is t_i, and the closest point of approach (CPA) relative bearing at the moment closed to point of approach is b_i. The collision threat layer $\boldsymbol{L}_c$ is defined as the relative position of all obstacles in the layer coordinate frame $\{l\}$ at the moment closed to point of approach, and $\hat{\boldsymbol{\eta}}_i = [d_i \cdot \sin(b_i), d_i \cdot \cos(b_i)]^{\mathrm{T}}$ represents the relative position of obstacles and USV at the moment closed to point of approach. According to the mapping relationship, the coordinates and radius of the ith obstacle's relative encounter

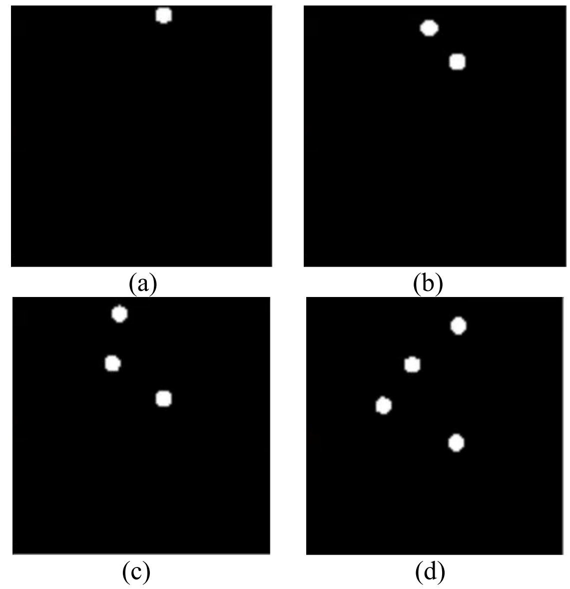

Fig. 5. Position layer image. (a) $t = 55$. (b) $t = 80$. (c) $t = 105$. (d) $t = 130$.

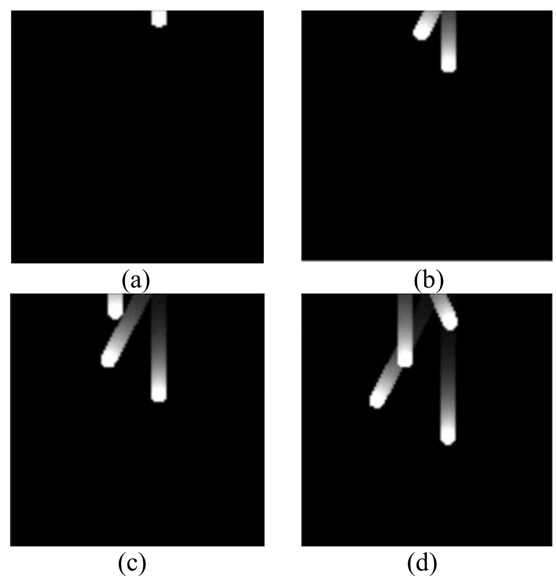

Fig. 6. Obstacle track layer image. (a) $t = 55$. (b) $t = 80$. (c) $t = 105$. (d) $t = 130$.

location feature in $\{l\}$ are $\boldsymbol{R}(\kappa)\hat{\boldsymbol{\eta}}_i$ and κr_i, respectively, and the intensity value of the coverage area of the ith obstacle's encounter point in $\boldsymbol{L}_c$ is

$$\begin{cases} 255 \times \gamma_c^{t_i}\ (t_i \geq 0) \\ 0\ (t_i < 0) \end{cases} \tag{13}$$

where γ_c is the threat decay coefficient. When $t_i \geq 0$, the relative distance between the USV and the obstacle gradually decreases. Therefore, the more t_i approaches 0, the collision threat is greater and the intensity value is closer to the maximum 255. When $t_i < 0$, the USV is far away from the obstacle, and the threat is considered eliminated, so the intensity value is set to 0. The encounter point in $\boldsymbol{L}_c$ can represent the threat level of collision between the obstacle and the USV. When the encounter point is close to the center of

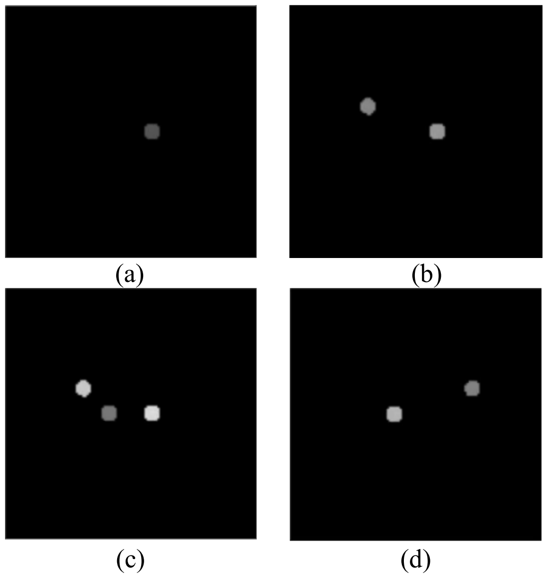

Fig. 7. Collision threat layer image. (a) $t = 55$. (b) $t = 80$. (c) $t = 105$. (d) $t = 130$.

the layer, it means that the USV will collide if it keeps the original course. The greater the intensity value of the encounter point is, the faster the collision will be happening and the more urgent the threat will be. Therefore, when there is no encounter point with a high-intensity value in the center of the layer, it can be considered that the collision threat of the USV is low.

Fig. 7 shows the collision threat layer corresponding to the four moments in the collision avoidance scenarios. As the USV remains a stable course throughout the whole process, the position of the encounter point in the layer remains unchanged, but the intensity will gradually increase with the decrease of the encounter time. It can be observed that there is a single encounter point in Fig. 7(a), and the intensity value of the encounter point in Fig. 7(b) and (c) gradually increases. Combined with Fig. 6, it can be found that the obstacle is opposite to the starboard side of the USV and the encounter has been completed at $t = 130$, so the encounter point in Fig. 7(d) disappears.

2) Design of Low-Dimensional State Feature: Although high-dimensional state features can represent obstacle situation information, the USV's navigation state and reference mission path are not suitable for 2-D image representation, so it is necessary to introduce low-dimensional state features as a supplement to high-dimensional state features.

The low-dimensional state in this article includes an LOS guidance error, the last time action output a_{t-1}, and the yaw rate $\dot{\psi}$. The LOS guidance error includes information about distance error y_e, $\dot{y}_e$ and heading error χ_d, $\dot{\chi}_d$.

In summary, the low-dimensional state feature vector $\boldsymbol{S}_l$ is designed as

$$\boldsymbol{S}_l = \left[y_e, \dot{y}_e, \chi_d, \dot{\chi}_d, a_{t-1}, \dot{\psi}\right]. \tag{14}$$

E. Design of Deep Network

In this article, a deep CNN fused with multifeature and multiscale is designed. The schematic of the network is shown in Fig. 8. First of all, the network inputs a double-channel image that combines the obstacle track layer $\boldsymbol{L}_s$ and collision threat layer $\boldsymbol{L}_c$, then the image is processed by the convolution-pooling layer several times. After this process, the dimension of the feature tensor is compressed from 117 × 117 to 3 × 3, where the number of channels is extended from 2 to 64, and then the feature tensor is flattened to the 1-D feature vector. Eventually, we will gain the public feature vector by concatenation of low-dimensional state features and the 1-D feature vector.

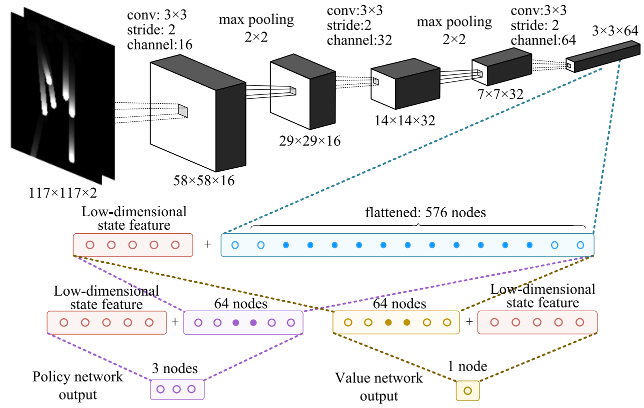

Fig. 8. Structure diagram of deep CNN fused with multifeature and multiscale.

The parameters of the network layer are shown in Table I. The PPO network consists of the policy network and the value network, which share common feature vectors and obtain corresponding outputs through two independent fully connected networks. After multiple convolution-pooling dimension reduction operations, there is a mismatch between the vector length of high-dimensional features (576) and low-dimensional state features (6). Consequently, low-dimensional state features are introduced into the input of the two fully connected networks to accelerate the convergence of the network.

IV. Network Training and Validation

A. Environment Construction and Training

Based on the OpenAI gym framework, the virtual environment of USV collision avoidance was constructed, which can support the visual display of training effects, and the training situation can be monitored and controlled by drawing collision avoidance scenarios in real-time. The training platform hardware was a workstation with i9-10980XE CPU, RTX2080Ti GPU, and 128 GB RAM. The training parameters were set as follows: The USV control period is 1 s, and after parameter tuning work, the adjustment coefficients k_1, k_2, k_3, and k_4 are set to be 1/100, 1/30, 5, and 1/50, respectively, the weight coefficients w_d, w_ψ, and w_s are 0.4, 0.4, and 0.2, respectively; reward value parameters r_o, r_c, and r_e are -500, -1000, and 50, respectively; the threat decay coefficient γ_s is 0.95 and $\Delta\psi_d$ is 10; the number of obstacles with high threat level is randomly selected between 1 and 2. The settings of the PPO network training hyperparameters are shown in Table II.

TABLE I
DEEP NETWORK PARAMETERS

Layer Seq.	Network Layer	Layer parameters	Activation function	Output dimension
1	input layer	——	——	117×117×2
2	conv 1	kernel 2×2, ch 16	leaky relu	58×58×16
3	pooling 1	2×2 max pooling	——	29×29×16
4	conv 2	kernel 2×2, ch 32	leaky relu	14×14×32
5	pooling 2	2×2 max pooling	——	7×7×32
6	conv 3	kernel 2×2, ch 64	leaky relu	3×3×64
7	flatten	——	——	6+576
8-1	policy FC1	582×64 FC	tanh	6+64
9-1	policy FC2	70×3 FC	tanh	3
8-2	value FC1	582×64 FC	tanh	6+64
9-2	value FC2	70×1 FC	tanh	1

TABLE II
SETTINGS OF HYPERPARAMETERS

Hyper-parameters	Value
training epochs N	10000
experience buffer size R	8192
episode length E	300
discount factor γ	0.99
clip ratio ε	0.2
learning rate l_r	0.0001
optimizer	Adam

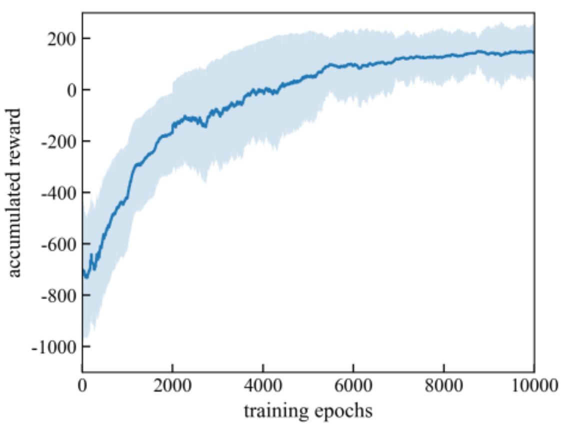

Fig. 9. Accumulated reward curve.

In this article, TensorFlow was used as the training framework for deep learning. During training, the high-dimensional state of network inputs $\boldsymbol{L}_s$ and $\boldsymbol{L}_c$ are normalized. The low-dimensional state feature vector $\boldsymbol{S}_l$ is standardized. At the end of each training epoch, sum the rewards of each time step in the current epoch to obtain the accumulated reward curve, which is shown in Fig. 9.

The blue line in Fig. 9 represents the moving average of accumulated reward (the sliding window is 100 epochs), and the shaded blue area represents the standard deviation of the moving average of accumulated reward. It can be seen that with the increase of the training epochs, the accumulated reward increases rapidly. After the training epochs reach 8000, the accumulated reward remains stable, indicating that the policy has been effectively converged.

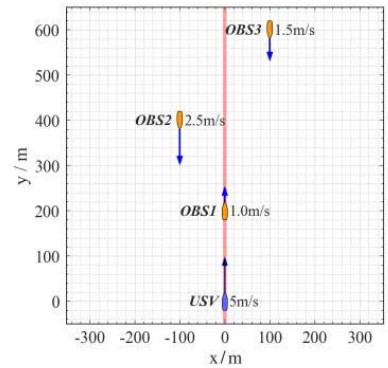

Fig. 10. Collision avoidance test scenario.

After the training, network parameters of different training epochs were used to test the collision avoidance effect of the USV, and the scenario of multiobstacles encounter as shown in Fig. 10 was established. In Fig. 10, the blue ship represents the USV, and the orange ships represent obstacles. The desired speed of the USV is 5 m/s, and the speed of OBS1, OBS2, and OBS3 are 1, 2.5, and 1.5 m/s, respectively. The red line in Fig. 10 is the reference path of the USV.

Fig. 11 shows the collision avoidance track of the USV in the test scenario at different training stages. In Fig. 11(a), the collision avoidance task is terminated in advance because the USV enters the collision threat area, and it is insufficient to learn the collision avoidance policy well after only 5 epochs of training. In Fig. 11(b), the USV can pass through obstacles with a zigzag track but fails to return to the reference course. It can be found that after 500 epochs of training, the USV has preliminarily acquired the ability to avoid moving obstacles. In Fig. 11(c), after 5000 epochs of training, the USV can avoid collisions and drive along the reference route at the same time, but the desired course output by the policy network is unstable, resulting in obvious fluctuation of the track of the USV. Fig. 11(d) shows the final training effect of the DRL network. It can be seen that the track fluctuation problem has been significantly improved, which indicates that the DRL algorithm proposed in this article can gradually master the collision avoidance policy through self-learning.

B. Dynamic Collision Avoidance Test

In order to further verify the effectiveness of the DRL collision avoidance policy, the method we proposed and the conventional method were used to simulate the collision avoidance situation at sea. In this article, two representative collision avoidance scenarios were selected to simulate the scenarios of USV driving in a busy port and crossing a busy channel, respectively, hereinafter referred to as Scenario 1 and Scenario 2. The conventional method is to use the APF method to avoid collisions.

1) Collision Avoidance Test in Scenario 1: Consider the collision avoidance scenario as shown in Fig. 12. In Scenario 1, five obstacle ships (OBS1–5) are defined as obstacles of the USV, and scenarios such as overtaking from the rear (OBS1),

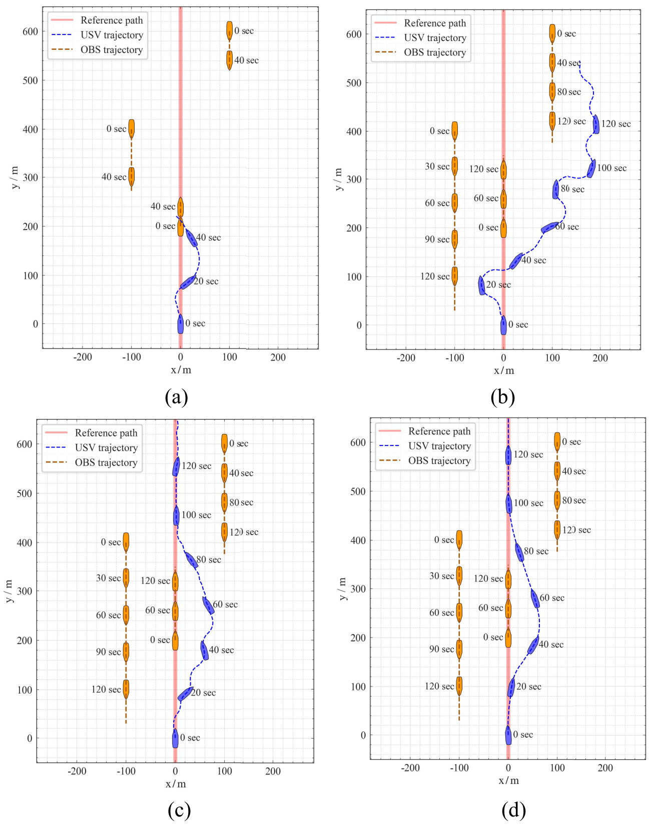

Fig. 11. USV collision avoidance track in different training periods. (a) 5 epochs. (b) 500 epochs. (c) 5000 epochs. (d) 10 000 epochs.

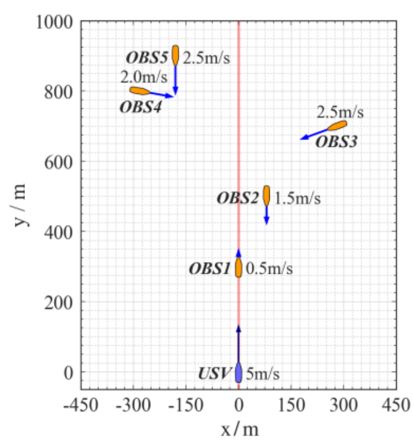

Fig. 12. Schematic of initial conditions in collision avoidance Scenario 1.

avoiding head-on encounter ships (OBS2 and OBS5), and avoiding crossing ships (OBS3 and OBS4) are simulated. First, in the case of USV overtaking the ship in front of it (OBS1), the collision of the meeting ship (OBS2 and OBS5) should be avoided at the same time. When overtaking is complete, the USV will begin to avoid crossing ships (OBS3 and OBS4). Finally, the USV should continue sailing on the reference path after completing a series of collision avoidance maneuvers.

Figs. 13 and 14 show the simulation results of multiple ship collision avoidance by the DRL method proposed in this article and the conventional method, respectively. On the left is the path of the USV to avoid collisions, and on

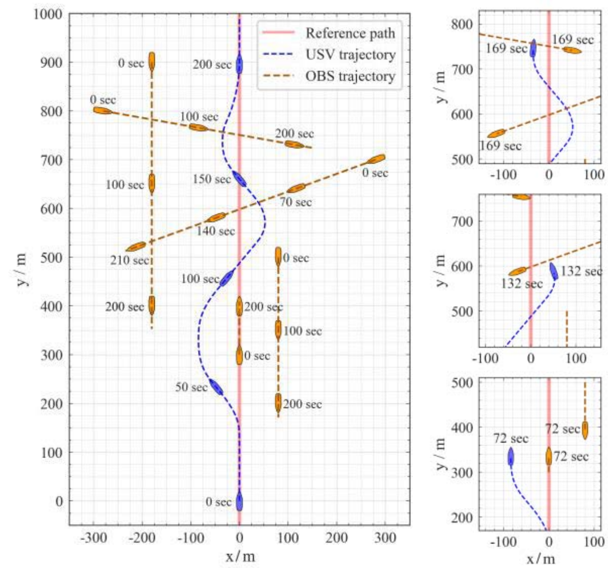

Fig. 13. Simulation result for multiple ship collision avoidance using DRL-based approach in Scenario 1.

the right is the situation when the USV evades different ships. When OBS1 is overtaken, both methods could consider the influence of the position of OBS2, and the obstacle is overtaken from the port side at 72 and 84 s, respectively. Compared with the conventional method, the path of the DRL method is smoother and less time-consuming. However, the conventional method turns sharply to the left after approaching OBS1 and then overtakes OBS1 after recovering to keep a safe distance. This is because, at the initial moment, the USV is subjected to both forward gravitational force and the repulsive force of OBS1 on the reference path. The two directions are opposite, and the direction of the resultant force is still forward. As the USV gradually approaches OBS2, the USV is affected by its repulsive force and turns to the port side. At this time, the component of the resultant force on the USV on the x-axis increases rapidly, leading to the appearance of a winding trajectory. When OBS3 and OBS4 are evaded, the path of the conventional method is similar to that of OBS1, and the trajectories swing violently. Compared with the conventional method, the DRL method can effectively learn collision avoidance policy in the training stage, which can understand collision environment information and make comprehensive decisions, so that it can complete collision avoidance tasks quickly and efficiently.

2) Collision Avoidance Test in Scenario 2: In collision avoidance Scenario 2 shown in Fig. 15, we set six obstacle ships. Each obstacle is numbered according to its initial distance from the USV from near to far. The obstacles are divided into two groups to simulate passing ships on both sides of the channel. Scenario 2 is designed to test the collision avoidance ability of the USV in the lateral encounter with a dense group of obstacles.

Figs. 16 and 17 show the simulation results of multiple ship collision avoidance by the DRL method proposed in this article and the conventional method, respectively. When OBS1-3 is avoided, the DRL method shows that the USV crosses the

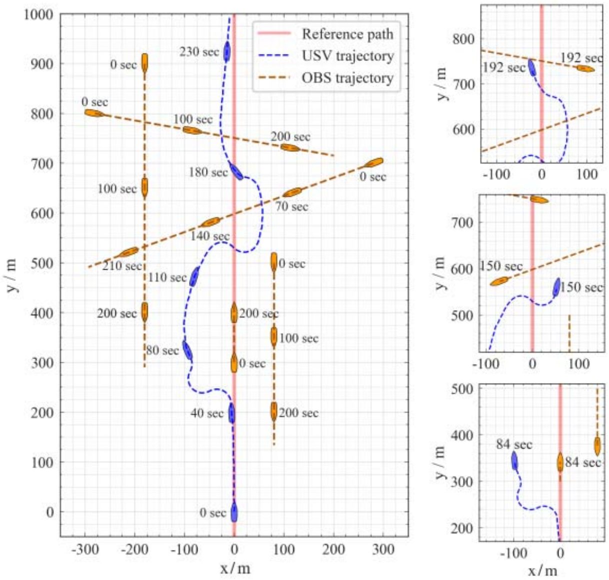

Fig. 14. Simulation result for multiple ship collision avoidance using APF-based approach in Scenario 1.

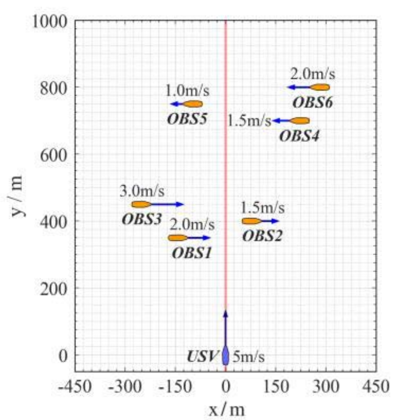

Fig. 15. Schematic of initial conditions in collision avoidance Scenario 2.

channel from the rear of the obstacle ship and successfully avoids OBS1 and OBS3 at 74 and 94 s, respectively. However, the conventional method crosses the channel from the front of the obstacle ship and avoids OBS1 and OBS3 at 83 and 105 s, respectively. This is because when the USV approaches OBS1, OBS1 is still on the port side of the USV, so the repulsive force generated by OBS1 will make the USV turn right. As time goes by, although the USV deviates from the reference path, under the influence of the repulsive force of OBS1, the USV has to keep the original course to ensure keeping a safe avoidance distance. In the latter part of the test, the DRL method could control the USV to cross the channel around the rear of OBS4, avoid OBS4 and OBS6 at 152 and 174 s, respectively, and quickly return to the reference path. The conventional method completes the avoidance at 160 and 181 s, respectively, but its speed of returning to the reference path is slower.

It can be found that the proposed DRL method is able to understand the state of motion and trends of collisions in the

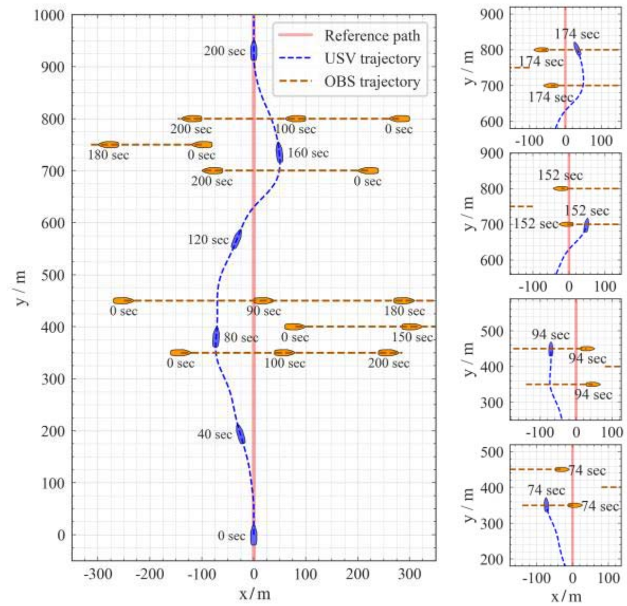

Fig. 16. Simulation result for multiple ship collision avoidance using DRL-based approach in Scenario 2.

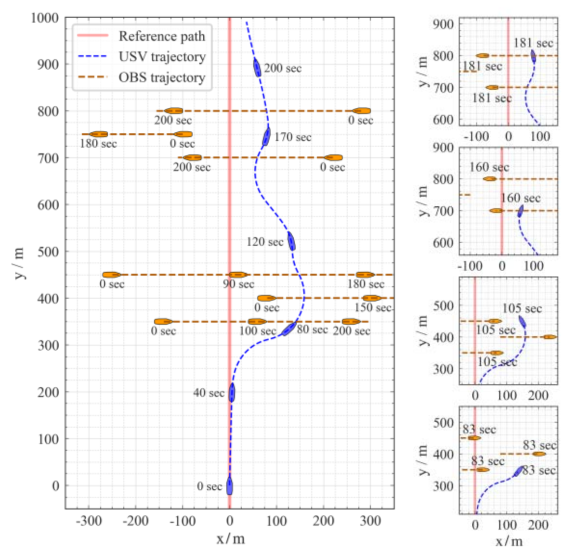

Fig. 17. Simulation result for multiple ship collision avoidance using APF-based approach in Scenario 2.

environment and make decisions consistent with long-term rewards. This capability benefits from the high-dimensional obstacle history tracking images and collision threat images fed to the DRL network, which enables it to make decisions based on the combined information of past, present, and future. However, conventional approaches in dealing with such obstacles will usually parameterize their situation, in which process most of the original information fails to be put to effective use. Therefore, the proposed DRL method can achieve better performance in complex collision avoidance scenarios.

C. Effectiveness Analysis

1) Ablation Experiment: To investigate the effects of the obstacle track layer, collision threat layer, and low-dimensional

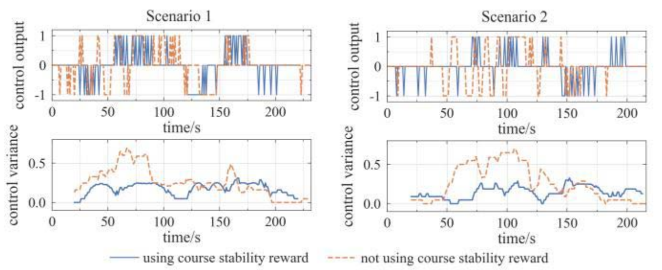

Fig. 18. Control output and variance comparison of different policies in the test scenarios.

TABLE III
RESULTS OF ABLATION EXPERIMENTS

Seq.	Ablation item	Averaged reward/step	Loss contribution
1	Baseline	0.608	0.00%
2	No obstacle track layer	0.486	-20.06%
3	No collision threat layer	0.579	-4.77%
4	No low-dimensional state	0.129	-78.79%

state, an ablation experiment is designed. Based on the network structure designed originally, the contribution of each function to the algorithm's performance is compared by masking certain structures. Averaged reward per step refers to dividing the sum of accumulated rewards by all the running steps in a random test environment rollout with a trained policy network. The experimental results are listed in Table III. It can be concluded that all of the designed states can improve the algorithm's performance, of which the low-dimensional state has the most notable contribution, and the obstacle track layer and collision threat layer also increase rewards by 20.06% and 4.77%, respectively. Ablation experiments have further verified the effectiveness of the proposed state design.

2) Course Stability Verification: To verify the effectiveness of the course stability reward, we remove it and retrain the obstacle avoidance policy network, which is tested in the two scenarios above. The output curves are shown in Fig. 18, where the solid blue and dashed orange lines are the original policy and the policy with the reward removed, respectively. The control output values 1, -1, and 0 denote increasing/decreasing the desired course and keeping the current course unchanged. By comparing the output curves, it can be observed that the policy output fluctuates after the reward is removed, which leads to the instability of the USV heading. The average output variances of the original policy and the removed policy over the last 20 time steps are 0.161 and 0.259, respectively, indicating that the addition of the course stability reward is beneficial for maintaining the stability of the USV course.

V. CONCLUSION

In this article, a DRL-based local collision avoidance algorithm for USV was proposed, and the effectiveness of the algorithm was validated by simulation. Through constructing the mathematical model of an underactuated unmanned vehicle, dynamic obstacle generation model and reward mechanism, the collision avoidance policy of the USV can be improved in the continuous interaction with the environment, so that it can make effective decisions in complex scenarios and complete the collision avoidance task quickly and efficiently. This article creatively designed a model of the end-to-end path planning algorithm and proposed two high-dimensional state features, including a layer that represents past and present obstacle tracks and a layer that predicts future events of collision threats. The deep CNN structure fused in multifeature and multiscale we designed can directly output decisions, which simplifies the conventional process based on rules of decision-making and has the potential for engineering application.

Although the DRL method proposed in this article has its advantages, there are also a few limitations. As the training process in this article is performed in the simulation environment, the influence of environmental factors has not been considered, and the influence of sensor observation error on model stability is ignored, which has a certain gap with the real sailing situation of the USV. In the future, the International Regulations for Preventing Collisions at Sea (COLREGs) will be introduced into the collision avoidance algorithm to adapt to a wider range of scenarios. In addition, extending the motion space to the speed control of the USV is also one of the future research directions.

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